$$K_P = \frac{(P_{AB})^2}{(P_{A_2})(P_{B_2})} = \frac{\left(\frac{2x}{2} \times P\right)^2}{\left(\frac{(1-x)}{2} \times P\right)\left(\frac{1-x}{2} \times P\right)}$$

$$K_P = \frac{4x^2}{(1-x)^2}$$

கொடுக்கப்பட்டவை $K_P = 1; \frac{4x^2}{(1-x)^2} = 1$

$\Rightarrow 4x^2 = (1-x)^2$

$\Rightarrow 4x^2 = 1 + x^2 - 2x$

$3x^2 + 2x - 1 = 0$

$$x = \frac{-2 \pm \sqrt{4 - 4 \times 3 \times -1}}{2(3)}$$

$$x = \frac{-2 \pm \sqrt{4+12}}{6}$$

$$= \frac{-2 \pm \sqrt{16}}{6}$$

$$= \frac{-2+4}{6}; \frac{-2-4}{6}$$

$$= \frac{2}{6}; \frac{-6}{6}$$

$x = 0.33; -1$ (not possible)

$\therefore [A_2]_{eq} = 1 - x = 1 - 0.33 = 0.67$

$[B_2]_{eq} = 1 - x = 1 - 0.33 = 0.67$

$[AB]_{eq} = 2x = 2 \times 0.33 = 0.66$

50. $K_{P_1} = 8.19 \times 10^2$ $T_1 = 298$ K

$K_{P_2} = 4.6 \times 10^{-1}$ $T_2 = 498$ K

$$\log\left(\frac{K_{P_2}}{K_{P_1}}\right) = \frac{\Delta H^o}{2.303\,R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)$$

$$\log\left(\frac{4.6 \times 10^{-1}}{8.19 \times 10^2}\right) = \frac{\Delta H^o}{2.303 \times 8.314}\left(\frac{498 - 298}{498 \times 298}\right)$$

$$\frac{-3.2505 \times 2.303 \times 8.314 \times 498 \times 298}{200} = \Delta H^o$$

$\Delta H^o = -46181$ J mol^{-1}

$\Delta H^o = -46.18$ KJ mol^{-1}

51. $P_{CO_2} = 1.017 \times 10^{-3}$ atm $T = 500^o$ C

$K_P = P_{CO_2}$

$\therefore K_{P_1} = 1.017 \times 10^{-3}$ $T = 500 + 273 = 773$ K

$K_{P_2} = ?$ $T = 600 + 273 = 873$ K

$\Delta H^o = 181$ KJ mol^{-1}

$$\log\left(\frac{K_{P_2}}{K_{P_1}}\right) = \frac{\Delta H^o}{2.303\,R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)$$

$$\log\left(\frac{K_{P_2}}{1.017 \times 10^{-3}}\right) = \frac{181 \times 10^3}{2.303 \times 8.314}\left(\frac{873 - 773}{873 \times 773}\right)$$

$$\log\left(\frac{K_{P_2}}{1.017 \times 10^{-3}}\right) = \frac{181 \times 10^3 \times 100}{2.303 \times 8.314 \times 873 \times 773}$$

$$\frac{K_{P_2}}{1.017 \times 10^{-3}} = \text{anti log of } (1.40)$$

$$\frac{K_{P_2}}{1.017 \times 10^{-3}} = 25.12$$

$\Rightarrow K_{P_2} = 25.12 \times 1.017 \times 10^{-3}$

$K_{P_2} = 25.54 \times 10^{-3}$

Unit - 9

Evaluate yourself :

1) KOH-ன் நிறை = 5.6 g

மோல்களின் எண்ணிக்கை $= \frac{5.6}{56} = 0.1$ மோல்

i) கரைசலில் கனவளவு = 500 ml = 0.5 L

ii) கரைசலில் கனவளவு = 1 L

$$\text{மோலாரிட்டி} = \frac{\text{கரைபொருள் மோல்களின் எண்ணிக்கை}}{\text{கரைசலில் கனஅளவு}}$$

$$= \frac{0.1}{0.5} = 0.2 \text{ M}$$

ii) = கரைசலில் கனஅளவு 1 L

மோலாரிட்டி $= \frac{0.1}{1} = 0.1$M

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